COMMERCIAL ITEM DESCRIPTION

TEST SET, TELEPHONE

The General Services Administration has authorized the use of this Commercial Item Description.

Scope. This Commercial Item Description (CID), describes a connector test set, hereafter called a CTS. This CID is meant as a minimum requirement for the CTS in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be type III, class 5, style B and color R in accordance with MIL-T-28800.

Applicable documents.

Specifications and standards. The following specifications and standards form a part of this CID to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DIDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for.

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Other publications. The following document forms a part of this CID to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI-C62.41-1980

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(Application for copies of ANSI publications should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

Order of precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards) the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Salient characteristics. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

1. Safety. The equipment shall comply with the type III safety requirements of MIL-T-28800.

a. Leakage current protection. Protection shall be provided from leakage current in excess of 3.5 milliamperes (ma) rms (AC and DC) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the lower source for any position of the power switch(es).

b. Radio frequency interference (RFI) capacitors. Any capacitor used in RFI filters shall be rated for at least four(4) times the operating voltage.

2. Parts, materials, and processes. Except for marking, the CTS shall comply with the type III parts, materials, and process requirements of MIL-T-28800. Nonstandard parts approval requests shall be in accordance with MIL-T-28800. Front panel marking shall be in accordance with MIL-T-28800.

a. Restricted material. The equipment shall comply with the restricted material requirements of the latest MIL-T-28800, including the requirement for mercury and radioactive materials, and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

3. Design and construction. Unless otherwise specified herein, the equipment shall comply with the type III design and construction requirements of MIL-T-28800. The equipment shall be of solid-state construction. Encapsulation and embedment (potting) of assemblies may be used only for high voltage circuits where construction is susceptible to leakage or corona, indicators and associated controls shall be in close proximity to each other, all potentiometer controls shall have at least a 270 degree rotation range (4.71 radians).

a. Modular design restrictions. The modular design concept for the CTS is acceptable with the following restrictions:

(1) Each modules plug-in card shall be keyed and mated with the back plane receptacle so that modules may not be randomly placed into the mainframe.

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(2) Modules shall not be removable by access through the front or rear of the mainframe without the unaided hand.

(3) A package part number shall be assigned to the CTS as a whole system. This system part number shall be marked on the mainframe identification plate and all plug-in(s).

(4) All module interface shall be done in the back of the mainframe.

b. Signal input and output connectors. Unless otherwise specified, all signal input and output connectors shall be 310-type connectors.

c. Internal cooling. The equipment shall be capable of operating continuously for twenty four(24) hours at 55°C. If forced air is used, it shall be as specified in MIL-T-28800.

4. Electrical power sources. The equipment shall operate from a central office battery of 48Vdc when test parameters required external power.

a. Surge and transient voltage protection. The CTS shall contain surge and transient voltage protection internal of the enclosure. The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system per type III solic state tolerance bands of MIL-T-28800, the surge voltage protection circuit shall conform to ANSI/IEEE standard C62.41-1980, 2.5KY, category A. The surge protection circuit shall drain excess power due to surges directly to the ground conductor for personnel protection. Normal operation of the CTS may be inhibited and false operational signals may be generated under input voltage transients greater than ±20 percent and frequency transient conditions greater than ±5 percent from within the nominal type III steady-state tolerance bands of MIL-T-28800, damage (other than fuse failure) to the CTS shall not occur during voltage transient conditions greater than ±20 percent and frequency transient conditions greater than ±5 percent within the maximum limits (30 percent voltage transients and 10 percent frequency transients) specified in MIL-T-28800. Normal operation of the CTS shall be automatically resumed when the transient has passed and any blown fuses are replaced. Blown fuses shall be accessible after removing front panel. Testing shall be in accordance with MIL-T-28800.

5. Dimensions and weight. The size and weight of individual component shall be consistent with current commercial capabilities, refer to paragraphs 5.a and 5.b of this CID.

a. Equipment dimensions. The nominal dimensions per component for the CTS should be approximately 8 inches high x 12 inches long x 9.5 inches deep.

b. Equipment weight. The weight including all accessories (excluding manual) per component, shall not exceed 20 lbs.

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6. Enclosure requirements. Unless otherwise specified herein, the equipment shall comply with the style B enclosure requirements of MIL-T-28800.

a. Hinged handle. A hinged handle shall be provided in accordance with MIL-T-28800. Exception to this paragraph may be acceptable as this is not a prime consideration of the equipment.

b. Carrying case. The CTS shall be supplied with a carrying case for protection when transporting or storing the CTS.

7. Marking and identification.

a. Marking. The visible surface adjacent to panel facilities such as variable controls, indicators, jacks, connectors and fuse holder(s), shall be marked with a suitable word, phrase, or abbreviation thereof, indicating the purpose or use of the part.

(1) Panel marking. The front panel marking shall comply with the type III parts, materials and processes panel marking requirements of MIL-T-28800, however, marking identification processes other than the type III requirements specified in MIL-T-28800 may be used, provided the marking and identification processes are capable of passing the front panel marking test when tested in accordance with this CID.

8. Environmental requirements.

a. Temperature requirements. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the class 5 temperature requirements specified by MIL-T-28800 and as listed below.

(1) Non-operating temperature. The equipment shall meet its performance and accuracy requirements in an non-operating environment of -51°C to +71°C.

(2) Operating temperature. The equipment shall met its performance and accuracy requirements in an operating environment of 0°C to 50°C.

b. Humidity requirements. The equipment shall meet its required performance and accuracy requirements under humidity requirements for class 5 equipment as specified by MIL-T-28800.

c. Altitude requirements. The equipment shall meet its required performance and accuracy requirements under the altitude requirements for class 5 equipment as specified by MIL-T-28800.

d. Vibration requirements. The equpment shall meet the class 5 vibration requirements as specified by MIL-T-28800.

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e. Mechanical shock requirements.

(1) Shock, functional. The equpment shall meet the class 5 shock, pulse requirement as specified by MIL-T-28800.

(2) Transit drop. The equipment shall meet transit drop requirement as specified by MIL-T-28800.

(3) Bench handling. The equipment shall meet the requirements as specified by MIL-T-28800.

f. Electromagnetic interference (EMI). The equipment shall comply with MIL-T-28800.

g. Magnetic environment. The equipment shall comply with the magnetic environment requirements of MIL-T-28800.

9. Reliability. Reliability verification testing, shall be in accordance with the type III, class 5 environmental conditions of MIL-T-28800.

a. Mean time between failures (MTBF). The equipment mean time between failures (MTBF) shall be 2,000 hours. The lower test MTBF shall be 2,000 and the upper test shall be MTBF 4,000.

b. Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

(1) Burn-in. Burn-in shall expose the equipment to extended periods of elevated temperatures, (40°C for a minimum of 100 hours). Each equipment which fails during burn-in shall be repaired and fully retested, until failure free. Failures which occur during the burn-in test shall be noted and reported but shall not count toward the establishment of MTBF. Prior to and following burn-in, a satisfactory operation test (SOT) shall be performed. A satisfactory operation check (SOC) shall be performed conducted at selected times during the burn-in test.

(2) Thermal cycling. Thermal cycling shall expose the equipment to hot and cold cycles for class 5 equipment per MIL-T-28800. Each equipment which fails during thermal cycling shall be repaired and full cycling test shall be noted and reported but shall not count toward the cycling, a satisfactory operation check (SOC) shall be conducted during the last two upper and lower temperature limits.

10. Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules.

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a. Maintainability. The equipment shall meet the type III maintainability requirements specified by MIL-T-28800.

11. Interchangeability. The equipment shall meet the type III interchangeability requirements specified by MIL-T-28800.

12. Performance characteristics.

a. The CTS shall be capable of performing operating and pulsing test on step-by-step connector switches. It must test local, toll, combination, level hunting and trunk hunting connectors.

b. The CTS shall perform loop and leak pulsing test on selector and connector switches.

c. The CTS shall automatically sequence through the following test series on idle lines:

(1) Pulsing test.

(2) Monitor ringing generator and sleeve ground test.

(3) Non-trip and trip test.

(4) Reversal test.

(5) Audible talk path test.

d. The CTS shall perform noid test for called and calling party.

e. The CTS shall also perform permanent signal test.

f. The CTS shall be equipped with a 1000Hz, 1mW power supply for performing the audible talk path test or for use in conjunction with a transmission measurement set.

g. The CTS shall be equipped with an auto dialer to out pulse preset digits for loop and leak test.

h. The CTS shall perform a busy line, leak A or B test.

i. Indicators. The CTS will be equipped with the following LED indicators on the front panel:

(1) Busy line - indicates connector in use.

(2) STR - indicates normal polarity of circuit.

(3) Seizure - indicates CTS has seized connector and is ready for test.

(4) Busy test number - indicates test number is made busy.

(5) "C" Lead ground - indicates ground has been applied to "C" lead.

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(6) Ringing Generator - indicates ringing generator is applied.

(7) Rev - indicates reverse polarity on circuit.

(8) Non-trip - indicates non-trip or ringing tripped.

(9) Trip - indicates trip test applied.

(10) Permanant - indicates permanent signal.

j. Test jacks. The CTS will be equipped with the following test jacks:

(1) Batt - connects central office (48V) battery and ground to CTS.

(2) 3R - connects the 3 jack (R in 335-type offices) of the connector group line jacks to the CTS.

(3) 4B - connects the 4 jack (B in 355-type offices) of the connector group line jacks to the CTS.

(4) T-R-S test number - connects the connector test number to the CTS.

(5) Level measuring - connects a transmission measurement test set to the CTS.

(6) T-R-S conn - connects the test jacks of a local connector to the CTS.

(7) L and R (twin plug) - connects the combination connector test jacks to the CTS.

(8) Permanant - connects the P jack of the connector group line jacks to the CTS.

k. Power and test cords. The CTS shall be equippec with the following power and test cords:

(1) (1) Power cord, 2P9A, 9 feet (2.74m), 2 conductors, 310-type plugs on both ends.

(2) (3) Line cords, 3P7A, 6 feet (1.8m), 3 conductors, 310-type plugs on both ends.

(3) (1) Connector cord, 10 feet (3.1m), 3 conductors, 310-type plug on one end alligator clips on the other end.

(4) (1) Test cord, 4P4A, 10 feet (3.1m), 4 conductors, 240B plug on one end and a 464C plug on the other end.

(5) (1) Test cord, 3P2A, 10 feet (3.1m), 3 conductors, 310-type plug on one end and a 240A plug on the other end.

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(6) Test cord, 4P5A, 12 feet (3.4m), 4 conductors, 240C plug on one end and a 264C plug on the other end.

13. Calibration interval. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

Preservation, packaging, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples. (The Government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Known acceptable products. The following product(s) are known to have previously met the requirements of this CID and need not resubmit bid samples or certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required.

Manufacturer	Model	NSN
CMC Telecom Corp	CMC-7810	6625-01-143-4628

Custodian
Air Force - 99

Preparing activity:
Air Force - 82

Review activitite:
Army - CR

User:
Navy - SH

(Project 6625-F894)

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